

ASSESSMENT GUIDE

ITEC631: Information Technology Masters Project Part B, Semester 2, 2026

Assessment Number 3 (A3) Research-Informed Final Project Thesis, Individual Reflection and Online Viva

Assessment key details

Assessment classification	Written, Oral, Experiential
Assessment method	Group project thesis, research and technical evidence, individual reflection and online viva
Assessment type	Lane-1 (Secure) and Lane-2 (Open)
Assessment artefact	Written thesis (2,000 words), research and technical appendices, final project package, individual reflection and online viva
Collaboration type	Group submission with a collective component and an individual component
Due date:	Week 12 - see Canvas for the exact date, time and viva schedule
Weighting:	50
Length and/or format:	2,000-word group thesis plus evidence appendices, individual reflection and online viva
Learning outcomes assessed	LO1; LO2; LO3; LO4
Graduate capabilities assessed	GC1, GC2, GC4, GC11, GC12
How to submit:	Thesis through Turnitin in Canvas; supporting project files through the designated Canvas submission area
Return of assignment:	Two weeks after submission
Assessment criteria:	Rubric: see the end of this document

Why this assessment?

This assessment brings together the full project and requires the team to demonstrate:

- how research and prior literature informed the project;
- a defensible and working software outcome;
- evidence from implementation, testing and evaluation; and
- individual ownership through contribution evidence, reflection and the online viva.

What employability skills will I develop through this assessment?

Skill Type	
Work readiness	☒
Knowledge of study areas	☒
Spoken communication	☒
Written communication	☒
Confidence to learn Independently	☒
Teamwork developed	☒
Complex problem solving	☒
Critical and analytical thinking	☒

Assessment overview:

Purpose:

Assessment 3 is the final and most substantial assessment in ITEC631. It brings together the research work begun in ITEC630, the software engineering work completed in ITEC631, and the evidence used to evaluate the final project. The thesis must show not only what the team built, but also what problem was investigated, how prior research informed the work, what evidence was collected, what was learned, and why the outcomes matter.

The assessment also verifies individual ownership. The group thesis and project evidence are supported by an individual contribution statement, an individual reflection and an online viva in which each student must explain and defend the work attributed to them.

Context

In ITEC630, students completed a literature review and project planning activities. That work provides the research foundation for this thesis. Students should not reproduce the earlier literature review. Instead, they must update and selectively synthesise the most relevant research, show how it informed the project, and connect it to the final design, implementation, evaluation and recommendations.

The published thesis length is 2,000 words. The main thesis must therefore be selective, analytical and evidence-based. Detailed literature matrices, requirements, diagrams, evaluation instruments, test records, data summaries, contribution logs and handover material should be organised in appendices and clearly cross-referenced from the main discussion.

Instructions

Submit one group thesis in Week 12. The Assessment 3 mark is divided equally: 50% is a collective group mark for the thesis, research and engineering evidence, and 50% is an individual mark based on contribution, reflection and performance in the online viva / presentation.

All students must participate in the viva / presentation. Questions may cover the research problem, literature, methodology, requirements, design, code, testing, evaluation, results, limitations, contribution evidence, use of AI and each student's professional learning.

Research and project requirements

- Research focus: state a clear project problem and one or more focused research or investigation questions. The questions must be relevant to the project and answerable using the evidence collected during development and evaluation.
- Updated literature synthesis: build on the ITEC630 literature review by selecting the most relevant academic and authoritative technical sources. Update the review where necessary and explain the research gap, practical problem or unresolved issue that the project addresses. Avoid a sequence of disconnected source summaries.
- Research contribution: explain the contribution made by the project. This may be a tested software artefact, an improved process, an evaluated design, evidence about user needs or system performance, or a practical insight that extends or challenges the earlier literature.

- **Methodology:** identify and justify an appropriate approach for the project. Depending on the project, this may involve design science, a case study, experimental or comparative evaluation, usability evaluation, mixed methods, or another suitable approach. Explain the steps followed, evidence collected, analysis used and why the approach was appropriate.
- **Software engineering:** systematically document the requirements, architecture, design, implementation, configuration, testing and evaluation needed to support the research and project objectives. Requirements, implementation and evidence must be traceable rather than presented as separate activities.
- **Results and discussion:** present findings using appropriate quantitative and/or qualitative evidence. Interpret the results against the research questions, success criteria, requirements and relevant literature. Where feasible, compare the final system with a baseline, previous process, alternative design or agreed target.
- **Limitations and validity:** identify unresolved technical issues, data limitations, sample or evaluation constraints, assumptions, threats to validity and circumstances in which the findings may not generalise. Do not present the project as more complete or conclusive than the evidence supports.
- **Research integrity and ethics:** use genuine evidence and maintain accurate records. Fabricated tests, invented feedback or altered results are not acceptable. Any user or stakeholder feedback must be collected under the approved class or project arrangements. Do not collect sensitive data or describe informal feedback as formal human-participant research without appropriate approval.
- **Common good and professional responsibility:** explain how the project affects people and organisations. Address the ethical, social, privacy, security, accessibility, fairness, sustainability or human-impact issues that are genuinely relevant to the project and show how they influenced decisions.
- **Technical handover:** the submission should allow another competent person to understand the system, review the evidence, set it up where practical, identify dependencies and known issues, and continue the work.
- **Individual contribution and reflection:** provide verifiable evidence of each member's research, technical and project contribution. Each student must include a concise reflection explaining responsibilities, decisions, challenges, learning, teamwork, ethical responsibilities and future professional development.
- **Originality and attribution:** acknowledge all external ideas, research, code, libraries, datasets, images, tools and AI assistance. The team remains responsible for the accuracy, legality, security, privacy and quality of the submitted work.

Required thesis structure

Title page and authorship statement: include the project title, team members, student numbers and a statement identifying the submitted work as the team's own, except where sources and assistance are acknowledged.

- **Executive summary (about 150 words):** provide a stand-alone summary of the problem, research question, approach, final artefact, main findings, limitations and recommendations.
- **Project context, research problem and common good (about 250 words):** explain the practical context, intended users or stakeholders, project aim, scope, research or investigation question(s), expected contribution and why the work matters.

Updated literature synthesis and gap (about 300 words): critically synthesise the most relevant research, prior systems, standards and technologies. Explain what was already known, what

remained unresolved, and how the evidence shaped the project direction. Do not copy the ITEC630 literature review.

Research and engineering methodology, requirements and design (about 350 words): justify the research or evaluation approach, development process, evidence sources, analysis method, requirements process, final scope, architecture and key design decisions. Refer to detailed protocols, models and traceability evidence in the appendices.

- Implementation and final artefact (about 250 words): explain the major components delivered, technologies used, integration work, important implementation decisions and how the artefact addresses the research and project objectives.
- Evaluation, results and discussion (about 450 words): present the most important test and evaluation results, including usability, performance, security, reliability or stakeholder evidence where relevant. Analyse what the results mean, answer the research question(s), compare the findings with the literature or agreed baseline, and distinguish evidence from interpretation.
- Project performance, limitations, conclusion and recommendations (about 250 words): compare the outcome with the plan and success criteria, identify major challenges and limitations, state the overall contribution, and provide justified recommendations for future development, deployment, maintenance or further research.
- References: use IEEE style and include only sources cited in the thesis. Cite borrowed diagrams, code, datasets, standards, frameworks and other external material.
- Appendices: provide organised evidence that supports the thesis. Appendices are not a substitute for analysis in the main text and should not contain large amounts of unedited or irrelevant material.
- Individual reflection and contribution evidence: include each student's reflection, the team contribution matrix and links or references to verifiable evidence. A maximum of 400 words per student is recommended for the reflection.

Process for completing the assessment

1. Review the ITEC630 literature review, proposal, approved scope, project plan, risk register and feedback. Identify which findings remain relevant and what needs to be updated for the final project. Each member have to review at least three papers and then synthesise all papers together into one form.
2. Confirm the project problem, research or investigation question(s), intended contribution and success criteria. Ensure that each question can be answered using available project evidence.
3. Update the literature selectively and prepare a literature-to-decision matrix showing how key sources informed requirements, design choices, implementation, evaluation criteria or interpretation of results.
4. Define and document the methodology and evaluation protocol. Specify what evidence will be collected, how it will be analysed, any baseline or comparison, and the limitations of the approach.
5. Prepare final requirements and research traceability: link each question and requirement to the relevant implementation, evaluation evidence, result and conclusion.
6. Consolidate and analyse genuine evidence, including representative test cases, defects and corrections, usability findings, performance or security results, stakeholder feedback and other project-specific measures.

7. Draft the thesis around a clear argument. Explain how the literature, methodology, artefact and evidence lead to the findings. Use tables and figures selectively and place detailed evidence in labelled appendices.
8. Prepare the technical handover package: final source-code snapshot or repository export, README, dependencies, setup instructions, test accounts where appropriate, known issues, deployment notes and data or evaluation documentation.
9. Prepare individual contribution statements and reflections. Complete peer review within the team and check the accuracy of research claims, results, references, diagrams, word count, confidentiality and removal of credentials.
10. Rehearse for the viva / presentation. Each student should be able to explain the research problem, one important source, the methodology, a major technical decision, a key result, a limitation, their own contribution and what they learned.

Suggested thesis and evidence structure

Part	Suggested focus	Evidence or content to include
1	Executive summary	Problem, research question, approach, artefact, key result, limitation and recommendation
2	Research problem and common good	Context, users, scope, question(s), intended contribution and relevant ethical or social impact
3	Updated literature and gap	Critical synthesis of relevant studies, systems, standards and the unresolved issue addressed
4	Research methodology	Chosen approach, evidence sources, evaluation protocol, analysis method, assumptions and limitations
5	Requirements and design	Final requirements, architecture, models, traceability and justification of key decisions
6	Implementation and artefact	Delivered components, technologies, integration, configuration and technical trade-offs
7	Evaluation and results	Test strategy, representative results, usability or other evaluation, defects and success measures
8	Discussion and contribution	Answer to the research question, comparison with literature or baseline, practical contribution and common good
9	Performance and limitations	Progress against plan, risks, unresolved issues, threats to validity and lessons learned
10	Conclusion and recommendations	Overall findings, recommendations for deployment, maintenance, development or further research
11	Individual evidence	Contribution matrix, individual reflections, peer assessment and repository or task evidence
12	Research and technical appendices	Literature matrix, research traceability, protocols, data summaries, models, tests, README, setup notes and AI disclosure

How do I submit?

- Upload the final thesis in Word or PDF format through Turnitin in the Assessment 3 area in Canvas. Follow the file format and naming instructions shown in Canvas.

- Upload the supporting project and research evidence package through the designated Canvas submission area. This should include the final code or repository export, README, required appendices, evaluation evidence and any other artefacts requested by the Lecturer in Charge.
- Include one consolidated contribution statement identifying each member's research, technical and writing responsibilities and the evidence supporting them, together with each student's individual reflection. Submit the peer-assessment form if requested in Canvas.
- Include the signed AI disclosure form and the required details of AI use. Keep relevant prompts, outputs and draft versions because they may be requested during marking or the viva.
- Remove passwords, API keys, private data, confidential information and identifiable user data before submission. Keep a complete backup of the thesis, code, appendices, evidence and submitted files.

Student submission checklist

Before submitting, our team has checked that:	Done
The main thesis follows the required structure and remains within the published 2,000-word limit.	☐
The project problem, research or investigation question(s) and intended contribution are explicit.	☐
The ITEC630 literature review has been updated and synthesised rather than copied.	☐
Relevant literature is used to justify requirements, design, evaluation and interpretation of results.	☐
The research and software engineering methodology is appropriate, transparent and justified.	☐
All data, test results and user or stakeholder feedback are genuine and accurately reported.	☐
Results answer the research question(s) and are discussed against requirements, success criteria and literature or a baseline.	☐
Research questions, requirements, implementation, evaluation evidence and conclusions are traceable.	☐
Limitations, assumptions, unresolved risks and threats to validity are stated honestly.	☐
Ethical, privacy, security, accessibility and common-good implications are addressed where relevant.	☐
Each member has provided verifiable research and technical contribution evidence and an individual reflection.	☐
All members are prepared to explain the whole project and defend their own work in the online viva.	☐

- IEEE citations and the reference list are complete, accurate and consistent. ☐
- External code, datasets, images, tools and AI assistance are acknowledged and verified. ☐
- Passwords, API keys, confidential information and identifiable data have been removed, and the team has retained a complete backup. ☐

Helpful resources

- The ITEC630 Assessment 2 literature review, marker feedback and search records.
- The approved project proposal, scope, research questions, requirements, project plan and risk register from ITEC630.
- Assessment 1 sprint-review feedback, Assessment 2 final-presentation feedback, the final repository, issue tracker, design models, test records and evaluation evidence.
- The ITEC631 unit materials on software engineering, research methods, testing and evaluation, together with the SFIA framework for professional skills.
- ACU Academic Skills Unit resources for research writing, synthesis, report structure, IEEE referencing and academic integrity.

Who can help me?

- Use the weekly workshop and consultation time to discuss the research framing, methodology, evaluation evidence, thesis structure and project-specific questions.
- Post general assessment questions in the ITEC631 Canvas discussion forum so that all students can benefit from the response.
- Contact the Lecturer in Charge at [walayat.hussain@acu.edu.au](mailto:walayathussain@acu.edu.au) for confidential matters, contribution disputes, research or ethics concerns, or questions specific to your team.
- Use the Academic Skills Unit for support with academic writing, literature synthesis, structure, paraphrasing and IEEE referencing.

I am having problems

Raise technical, research, team, authorship or contribution problems as early as possible. A concern reported only after submission may be difficult to verify or resolve fairly.

The automatic 72-hour extension may not apply to every component of a group assessment or scheduled viva. Check the current information in Canvas and ACU policy. If exceptional circumstances affect your work, follow the Special Consideration process and provide the required documentation.

Special Considerations

Refer to the Special Consideration procedure if your studies have been affected by unforeseen or exceptional circumstances and you are seeking exemption from academic penalty. [Special Consideration procedure](#)

Complete a Special Consideration application if: [Special Consideration application](#)

- you were unable to apply for an extension before the due date/time, and
- you have valid supporting reasons and documentation.

Special Consideration is different from the automatic three-day extension and is typically used for more significant or unexpected circumstances.

Referencing

Use IEEE referencing. Provide numbered in-text citations and a complete reference list. Cite research literature, external code, libraries, datasets, standards, diagrams, images and other material where relevant. Missing acknowledgement may constitute plagiarism or misrepresentation.

Academic integrity

Submit only work produced by your team and acknowledge the work of others. Copying code, diagrams, written material or project artefacts without appropriate acknowledgement may constitute academic misconduct. Each student must be able to explain the work attributed to them and demonstrate an understanding of the submitted system.

Please refer to the [Academic Integrity and Misconduct Policy and the Academic Misconduct Procedures](#) for full details. In particular, academic misconduct, including cheating, plagiarism, collusion, recycling of assignments, and misrepresentation, is not acceptable. Penalties for academic misconduct may vary in severity and can include exclusion from the course.

ACU takes academic misconduct very seriously. To ensure all students prepare their work with integrity, the LIC will contact you and ask you to explain your work and how you prepared it, should they suspect that academic misconduct may have occurred.

Use of generative AI

Generative AI tools may be used in a limited and transparent way to support learning and development. Appropriate uses may include brainstorming, explaining concepts, identifying possible coding errors, suggesting test cases, improving wording, or reviewing the clarity of documentation.

AI output must not replace the team's own analysis, design decisions, implementation and understanding. Students remain responsible for checking the accuracy, security, privacy, licensing and suitability of any AI-assisted output. Do not enter confidential project, client or personal data into a public AI system.

AI disclosure requirements

Upload the AI disclosure form signed by all team members.

- State whether AI tools were used.
- Name the tool and explain the purpose for which it was used.
- Identify the slides, code or project artefacts that were influenced by AI assistance.
- Explain how the output was checked, changed and integrated by the team.
- Keep relevant prompts and outputs. These may be requested if clarification is required.
- Be prepared to explain any AI-assisted code or content during the presentation and questions.

Important

Assessment 3 is not a second literature review and it is not only a software manual. It must show what the team investigated, what the literature indicated, what was built, how it was evaluated, what the evidence means, and what was learned.

Seeking clarification

If you are unsure whether an AI tool, research activity or another form of assistance is acceptable for a particular task, contact the Lecturer in Charge before proceeding.

Rubric criteria

The rubric separates the collective quality of the group thesis and supporting evidence from each student's individual research and technical contribution, reflection and viva performance. The full criteria appear on the following pages.

Optional student self-assessment and feedback request proforma

All students will receive feedback based on the rubric. If the team would like more specific feedback, complete the table below and submit it with the thesis. Focus on questions that will help you understand the quality of the research, engineering work and final evidence.

Questions	Reference Criterion	Team's Response
What is the strongest evidence that the project answers its research question or achieves its intended contribution?		
Which research, technical or evaluation claim is least convincing?		
What specific feedback would help us improve the rigour or clarity of the thesis?		
How did we use feedback from ITEC630, Assessment 1 and Assessment 2?		

Feed Forward Template: Using feedback from the project

Review the feedback received in ITEC630, Assessment 1 and Assessment 2. Explain how it was used to improve the research framing, final system, evaluation and thesis, and identify advice that would support future professional or research work.

https://acu.qualtrics.com/jfe/form/SV_6QBbEb9U45gfUdU

Example Questions	Feedback recommendation (what we learned and how we used it)
1. How did earlier feedback change the research question, system, evaluation or thesis?	Identify the feedback, action taken and evidence that the change improved the work.
2. Which research, technical or professional skill needs further development after this unit?	Use a specific project example and identify a practical next step.
3. Was any feedback difficult to understand or apply?	Identify the unclear point and explain what additional guidance would have helped.

Rubric: Assessment 3 - Research-Informed Final Project Thesis, Individual Reflection and Online Viva (50%)

Criterion	Marks	Below expectations	Meets expectations	Credit	Distinction	High Distinction
		NN 0-49%	PA 50-64%	CR 65-74%	DI 75-84%	HD 85-100%
1. Research problem, literature synthesis, contribution and common good Group component GC1 / LO1	6	The research problem or question is unclear, unsupported or disconnected from the project. Literature is absent, inaccurate or largely descriptive. The contribution and common-good relevance are not established.	Identifies a basic problem or question and relevant literature, but synthesis, gap, contribution or common-good connection is limited.	Presents a clear research problem and credible synthesis of relevant literature. Identifies an appropriate gap or practical issue, intended contribution and common-good considerations.	Provides a well-focused and critical synthesis that positions the project convincingly. The research question, contribution and ethical or social significance clearly inform project decisions.	Provides an insightful and authoritative research framing. Literature is critically integrated to establish a compelling question, defensible contribution and sophisticated understanding of the common good.
2. Research and software engineering methodology, requirements, design and implementation Group component GC2 / GC4 / LO2 / LO3	7	The methodology and engineering process are poorly documented or inappropriate. Requirements, design and implementation are incomplete, inconsistent or unsupported.	Describes a basic methodology and the main technical elements, but justification, traceability and links between research and engineering work are limited.	Uses an appropriate research and engineering approach. Requirements, architecture and implementation are credible, and important methods and decisions are explained.	Presents a coherent, transparent and traceable methodology. Research evidence, requirements, design and implementation are strongly connected, and important trade-offs are justified.	Presents a rigorous and highly coherent methodology with advanced technical judgement. The research and engineering process is reproducible, fully traceable and convincingly aligned to the project questions and objectives.
3. Evaluation, results, discussion, limitations and recommendations Group component GC1 / GC2 / LO1 / LO2 / LO3	8	Evaluation is absent, superficial or unreliable. Results do not answer the research question or support success claims. Discussion, limitations and recommendations are inadequate.	Provides basic testing or evaluation and identifies some results, but analysis, comparison, interpretation or acknowledgement of limitations is limited.	Uses appropriate evidence to assess the artefact and answer the main question. Results are interpreted against requirements or success criteria, with realistic limitations and recommendations.	Provides systematic evaluation and strong critical analysis. Findings are compared with literature, a baseline or agreed targets, and limitations, validity concerns and recommendations are well justified.	Provides rigorous and compelling evaluation with precise interpretation of quantitative and/or qualitative evidence. Conclusions are defensible, limitations are handled maturely, and the recommendations make a significant practical or research contribution.
4. Thesis quality, evidence, traceability and technical handover Group component GC2 / GC11 / LO2 / LO3	4	The thesis is unclear, poorly organised or substantially incomplete. Evidence, references, appendices or handover material are inadequate, and claims cannot be independently followed.	The thesis is understandable and contains the main elements, although structure, concision, referencing, cross-referencing or supporting evidence is uneven.	The thesis is clear, concise and professionally structured. Figures, references and appendices support the discussion, and research and technical evidence can be followed.	The thesis communicates complex research and engineering work effectively. Evidence is well selected and cross-referenced, and the handover package is complete and professional.	The thesis is exceptionally clear, economical and persuasive. Evidence, traceability and handover are exemplary, allowing the research process, technical decisions, results and limitations to be independently examined.

Criterion	Marks	Below expectations	Meets expectations	Credit	Distinction	High Distinction
		NN 0-49%	PA 50-64%	CR 65-74%	DI 75-84%	HD 85-100%
5. Individual research and technical contribution, ownership and teamwork Individual component GC4 / LO3	10	Contribution is unclear, limited or unsupported. The student cannot adequately explain the research or technical work attributed to them or their role in the team.	Provides basic evidence of contribution and can explain some assigned work. Understanding of the research context, wider system or team integration is limited.	Provides credible evidence of substantial research and/or technical contribution, explains responsibilities and demonstrates effective teamwork and ownership.	Provides strong and well-organised evidence of contribution. The student explains complex research and technical work confidently, supports team integration and shows sound professional judgement.	Provides compelling evidence of an exceptional contribution and deep ownership. The student demonstrates leadership, dependable teamwork and sophisticated understanding of how their work shaped the research and project outcomes.
6. Individual critical reflection, research integrity and professional development Individual component GC1 / GC4 / LO1 / LO3	7	Reflection is descriptive, generic or unsupported. Research integrity, challenges, learning, ethical responsibilities and professional development are not meaningfully examined.	Identifies some learning, challenges and skills, with a basic connection to research integrity and professional practice.	Provides a clear reflection supported by project examples, including important challenges, decisions, research or technical learning, teamwork and areas for further development.	Provides a thoughtful and critical reflection on evidence, setbacks, teamwork, ethical responsibilities, research integrity and professional growth, supported by specific examples.	Provides an insightful, candid and evidence-based reflection demonstrating mature professional identity, strong research integrity, exceptional self-awareness and a convincing plan for continuing development.
7. Online viva: explanation, defence and response to questions Individual component GC12 / LO4	8	The student has difficulty explaining the research, project or their own work, gives inaccurate answers, or cannot defend key decisions, evidence and conclusions.	Explains the main aspects of their work and answers straightforward questions, but responses may be uncertain, descriptive or weakly supported.	Communicates clearly to specialist and non-specialist audiences, explains their contribution accurately and answers questions using appropriate research and project evidence.	Provides confident, well-reasoned and technically sound responses. The student defends the methodology and decisions, interprets results, acknowledges limitations and adapts explanations to the audience.	Provides precise, authoritative and reflective responses demonstrating deep ownership, advanced research and technical judgement, and an exceptional ability to communicate and defend complex work.
Total	50					